

Chapter 74 — Disorders of Male Orgasm and Ejaculation

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- **Ejaculatory dysfunction (EjD) is the most common male sexual disorder.**
- **Spectrum:** premature ejaculation (PE) → delayed ejaculation (DE) → complete inability to ejaculate (**anejaculation**); also retrograde ejaculation (RE), painful ejaculation, ejaculatory anhedonia and the recently described **postorgasmic illness syndrome (POIS)**.
- **Sexual response cycle** (Masters and Johnson): desire → arousal → orgasm → resolution. Arousal rising to a **threshold triggers the ejaculatory response**, which usually terminates the male sexual episode.
- **Orgasm is a distinct cortical event** — the perception, via pelvic sensory neurons, of striated muscle contraction and semen expulsion, experienced both cognitively and emotionally.
- **Ejaculatory latency is a continuum** (onset of penile stimulation → ejaculation), varying between and within men. Most ejaculate after several minutes and are satisfied; some ejaculate very rapidly, even before penetration; others only with great difficulty or not at all.
- **RE:** pharmacological, anatomical or neurological factors that **prevent bladder neck contraction**.
- **Painful ejaculation** (dysejaculation, odynorgasmia, postorgasmic pain, dysorgasmia, orgasmalgia) is prevalent and **commonly misdiagnosed and mistreated**.
- **Haemospermia** (blood in semen) alarms patients but is **typically benign and self-limiting**.

THE ANATOMY AND PHYSIOLOGY OF THE EJACULATORY RESPONSE

- **Reflex arc:** sensory receptors and areas → afferent pathways → cerebral sensory areas → cerebral motor centres → spinal motor centres → efferent pathways.
- The brain circuitry of ejaculation is **part of a wider network** controlling the whole sexual response.
- **Neurochemistry:** a complex interplay of **central serotonergic and dopaminergic neurons**, with secondary cholinergic, adrenergic, oxytocinergic and gamma-aminobutyric acid (GABA) involvement.
- **Periphery:** synergistic activation of **autonomic (sympathetic and parasympathetic) and somatic** divisions; nonadrenergic noncholinergic (NANC) innervation may modulate the accessory sex glands.
- **Five groups of spinal nuclei** (thoracolumbar and lumbosacral) give the autonomic and somatic efferents; their coordination is by lumbar interneurons at L3–L5 — the **spinal generator of ejaculation**. This spinal network is strongly influenced by stimulating or inhibiting **genital sensory and supraspinal inputs**.
- **Brain nuclei activated during ejaculation:** posteromedial bed nucleus of the stria terminalis, posterodorsal medial amygdala, posterodorsal preoptic nucleus and parvicellular subparafascicular thalamus.

Figure Legend

- **Fig. 74.1 — central nervous system network controlling ejaculation** (Clement and Giuliano, 2016): brain nuclei — posteromedial bed nucleus of the stria terminalis, posterodorsal medial amygdala, posterodorsal preoptic nucleus, parvicellular subparafascicular nucleus, medial preoptic area, paraventricular nucleus, lateral hypothalamus, gigantocellular nuclei and raphe — linked to spinal nuclei (sympathetic T12–L1, spinal generator of ejaculation drawn at L3–L4, sacral outflow S1–S3), with somatosensory afferents from the genital tract.

The Three Phases of Ejaculation

PHASES OF THE EJACULATORY PROCESS

Phase	Events	Mediation
Emission	Contraction of vas deferens, seminal vesicles and prostate → sperm and seminal fluid into the posterior urethra	Synergic sympathetic and parasympathetic , T10–L2; possibly a cholinergic excitatory mechanism
Ejection (penile expulsion)	Pulsatile bulbospongiosus, ischiocavernosus and levator ani contraction; bladder neck closes (prevents retrograde flow); external sphincter relaxes , with intense contractions interrupted by silent periods	Somatic S2–S4 via the pudendal nerve, plus a sympathetic spinal reflex under only limited voluntary control
Orgasm	Cerebral processing of pudendal sensory input from raised posterior urethral pressure, the verumontanum, and contraction of the urethral bulb and accessory sex organs	Cortical

- The table tests **which division drives which phase**: emission is autonomic, ejection somatic, orgasm cortical — hence ejaculation and orgasm can dissociate.

Neurotransmitters

- **Transmitters involved**: dopamine, noradrenaline, serotonin (5-hydroxytryptamine, 5-HT), acetylcholine, oxytocin, GABA and nitric oxide (NO); **dopamine and serotonin are the essential two**.

Dopamine	Serotonin
Promotes seminal emission and ejaculation via D2 receptors	Inhibitory ; neurons chiefly in brainstem, raphe nuclei and reticular formation

- **Receptor pair (rat data)**: 5-HT_{2C} agonism → **delays** ejaculation; postsynaptic 5-HT_{1A} stimulation → **shortens** ejaculation latency.
- Hence the hypothesis that men with PE have **5-HT_{2C} hyposensitivity and/or 5-HT_{1A} hypersensitivity**.

Concept — Ejaculation is a spinal reflex under supraspinal control whose set-point is serotonergic: **serotonin brakes and dopamine accelerates**, which is why PE drugs raise synaptic serotonin and DE drugs are dopaminergic or anti-serotonergic.

PREMATURE EJACULATION

Classification of Premature Ejaculation

- **Historical types B and A (1943)** → renamed (1989) **lifelong (primary) and acquired (secondary)**; later descriptors: global versus situational, substance-related.
- **Lifelong PE**: early ejaculation at **nearly every intercourse** — mostly **30–60 s**, a minority 1–2 min — with every partner, from the first encounters.
- **Acquired PE**: develops after previously normal ejaculation, often situational; distinguished by **time of onset** and **reduction from a previously normal latency**.
- **Normative intravaginal ejaculatory latency time (IELT) data**: IELT under 1 min is rare, yet many men with normal IELT complain of PE → the **Waldinger-Schweitzer four-subtype classification**, by IELT, complaint frequency and course in life.
- **Confirmed** in Turkish and Chinese cohorts: the complaint of PE affects roughly **one-fifth to one-quarter** of sexually active men; subtype order **variable > subjective > acquired > lifelong**.
- **Shared dimensions** of lifelong and acquired PE (lack of control, negative personal consequences) justified a **single unifying definition**, possibly to be extended later to variable and subjective PE.

COMPARISON — THE FOUR PE SUBTYPES (WALDINGER AND SCHWEITZER)

	Lifelong	Acquired	Variable	Subjective
Onset	First encounters	Later, after normal latency	—	—
Latency	≤ about 1 min (some 1–2 min)	Markedly reduced, often ≤ about 3 min	Occasionally short	Normal or long (2–6 min, up to 25)
Pattern	Every/nearly every time, every partner	Consistent; often situational	Occasional, irregular	Perceived
Basis	Neurobiological , genetic	Medical /psychological	Natural variation	Psychological/cultural
Disorder?	Yes	Yes	No	No
Setting	Dominate treatment-seekers		Dominate the community	

Concept — PE is classified on two axes, **objective latency and subjective complaint**; only where both are abnormal (lifelong, acquired) is there a disorder — variable and subjective PE are complaints without pathology.

Definition of Premature Ejaculation

- **Research depends on the definition used.** The literature holds many univariate and multivariate definitions, each built from some of: **ejaculatory latency, perceived control, reduced sexual satisfaction, personal distress, partner distress, interpersonal distress.**
- All the earlier definitions were **authority-based** — none supported by evidence-based clinical research.

TABLE 74.1 THE DEFINITIONS OF PREMATURE EJACULATION — AUTHORITY-BASED DEFINITIONS

Source	Latency element	Control / other
Masters and Johnson, 1970	None — partner-referenced	Cannot control long enough to satisfy partner in ≥50% of coital connections
ICD-10, 1994	Before or very soon after start; ≤15 s if a time limit is needed	Inability to control for both partners' enjoyment; or ejaculation without an erection adequate for intercourse; not due to prolonged abstinence
DSM-IV-TR, 2000	Minimal stimulation, before, on or shortly after penetration	Before the person wishes; marked distress or interpersonal difficulty; not solely a substance effect
EAU, 2001	Not for a "sufficient" length of time before penetration	Inability to control; no fertility impairment if intravaginal
Metz and McCarthy, 2003	—	No voluntary, conscious control or ability to choose when to ejaculate in most encounters
AUA, 2004	Sooner than desired, before or shortly after penetration	Distress to one or both partners
ICUD, 2004	Minimal stimulation, before, on or shortly after penetration	Little or no voluntary control; bother or distress to sufferer and/or partner
Jannini, 2005	Pathological IELT by stopwatch	Loss of control and/or distress or relational disturbance by PRO
Waldinger, 2005	<1 min (0.5th percentile) = definite ; 1–1.5 min (0.5th–2.5th) = probable	Further psychological grading: asymptomatic, mild, moderate, severe

ICD-10, *International Classification of Diseases, 10th edition*; DSM-IV-TR, *Diagnostic and Statistical Manual of Mental Disorders, 4th edition, text revision*; EAU, *European Association of Urology*; AUA, *American Urological Association*; ICUD,

International Consultation on Urological Diseases; PRO, patient-reported outcome. Contemporary definitions (ISSM 2008 and 2014, DSM-5 2013, AUA/SMSNA 2020) are in the comparison table below.

- The table traces the move from **qualitative or partner-referenced** definitions to **latency-anchored multidimensional** ones.
- **Pattern:** latency appears only as vague phrases until Waldinger's percentiles; only ICD-10 names a number (15 s).
- **Trap:** the 15-s figure is ICD-10's; Masters and Johnson make the diagnosis depend on the partner's response.

The International Society for Sexual Medicine Definition of Premature Ejaculation

- Enabled by the objective **IELT** plus validated **patient-reported outcome (PRO)** measures.
- **First International Society for Sexual Medicine (ISSM) ad hoc committee, October 2007** → first evidence-based definition of **lifelong PE** (2008). All three criteria required: (1) ejaculation **always or nearly always before or within about 1 min** of vaginal penetration; (2) **inability to delay** on all or nearly all penetrations; (3) **negative personal consequences** — distress, bother, frustration, avoidance of intimacy.
- The committee could **not** define acquired PE — insufficient data.
- **Second committee, Bangalore, April 2013:** lifelong and acquired PE are distinct populations but share the three constructs → **unified definition**; the added key dimension of acquired PE is a **clinically significant and bothersome reduction in latency, often to $\leq$ about 3 min**.
- First evidence-based definition of both; the **basis for office diagnosis** and for trial design.
- **Limited to vaginal intercourse** — few data in men who have sex with men or other sexual expression.
- **Deliberate conservatism with flexibility:** the 1-min cut-off is not absolute (a minority of treatment-seeking lifelong PE have IELT 1–2 min); "about 1 minute" allows diagnosis up to **90 s**; the acquired ≤ 3 -min threshold is applied with clinical judgement.
- **Latency alone never suffices:** short latency with adequate control and no negative consequences is **not** PE.

Diagnostic and Statistical Manual of Mental Disorders Definition of Premature Ejaculation

- Built on the same data as ISSM; the Diagnostic and Statistical Manual of Mental Disorders, 5th edition (DSM-5, 2013) now has an **objective latency criterion**: ejaculation in partnered activity **within approximately 1 min** of vaginal penetration and before the individual wishes, for **≥ 6 months**, on **75–100% of occasions**, with clinically significant distress.
- Specifiers: **lifelong or acquired; generalised or situational**.
- **Severity:** mild 30 s–1 min; moderate 15–30 s; severe before or at the start of sexual activity, or < 15 s.

American Urological Association/Sexual Medicine Society of North America Definition of Premature Ejaculation

- Disorders of Ejaculation panel formed **2018** by the AUA with the Sexual Medicine Society of North America (SMSNA); systematic review with an independent methodology team.
- **Critique of ISSM:** a substantial minority of treatment-seekers ejaculate after 60 s; the cut-off based on two standard deviations from the mean latency is statistically valid but **arbitrary, unnecessarily conservative and clinically dubious** as regards bother and control; the original data lacked non-PE comparison groups and independent criterion validation.
- **Supporting data:** men selected on poor control have median and mean latencies above 1 min; men at **60–120 s report distress like those under 60 s**, but unlike those over 120 s.

- **Lifelong PE (AUA):** poor control + bother + ejaculation **within about 2 min** of initiation of penetrative sex, present **since sexual debut**. Extending to 120 s raises true positives (fewer false negatives, type 2 error) at the cost of more false positives (type 1 error). Graded **expert opinion**.
- **Acquired PE (AUA):** consistently poor control + bother + latency **markedly reduced from prior experience** — **no latency figure in the statement**. Suggested guides: average latency under about **2–3 min**, or reduction of **≥50%** from before; acknowledged to lack evidence.

COMPARISON — CONTEMPORARY DEFINITIONS OF PE

	ISSM (2008; unified 2014)	DSM-5 (2013)	AUA/SMSNA (2020)
Lifelong latency	≤ about 1 min	≈ 1 min	≈ 2 min
Acquired latency	Bothersome fall, often ≤ 3 min	Same 1-min criterion; specifier only	None stated (guide 2–3 min or ≥50% fall)
Control	Inability to delay	Before one wishes	Poor control
Consequences	Negative personal consequences	Clinically significant distress	Bother
Duration / frequency	Always or nearly always	≥ 6 months; 75–100%	Since debut (lifelong)
Severity grades	No	Mild / moderate / severe	No
Evidence base	Evidence-based	Same data as ISSM	Expert opinion

Concept — Every contemporary definition is the same **three-construct test** — **latency, control, consequences**; the bodies differ only in where they draw the latency line, and the AUA's 2-min line trades specificity for sensitivity.

Prevalence of Premature Ejaculation

- Community estimates vary widely; PE is often called the **most common self-reported male sexual disorder**.
- **Disparity** between self-report or poorly validated definitions and stopwatch data.
- **Community stopwatch IELT:** distribution **positively skewed, median 5.4 min** (range 0.55–44.1 min), **falls with age**, varies between countries; IELT ≤1 min is **statistically abnormal**.
- **Self-report prevalence is appreciably higher** than clinician diagnosis using the conservative ISSM definition.
- **Global Study of Sexual Attitudes and Behaviours (GSSAB)**, age 40–80: high self-reported prevalence; perception of "normal" latency varied by country and between patient and partner; **limitation** — **no men under 40**. The **Premature Ejaculation Prevalence and Attitude (PEPA)** survey (age 18–70) likewise found high self-reported prevalence.
- **Treatment-seekers:** series differ on whether lifelong or acquired predominates, but **lifelong and acquired PE make up most patients seeking treatment**, whereas **variable and subjective PE make up most of the community** and seldom seek treatment.
- **Acquired PE** is uncommon in the community but its sufferers are **more likely to seek (and plan to seek) treatment** than those with lifelong PE.

Figure Legend

- **Fig. 74.2 — distribution of IELT in a random male cohort** (Waldinger, Quinn et al., 2005): a positively skewed curve — most men cluster at a few minutes with a long right tail.

Concept — The prevalence of PE is largely **an artefact of how it is defined**: self-report yields large figures, stopwatch-anchored definitions small ones, and the clinic population is enriched for the two true disorders.

Aetiology of Premature Ejaculation

- Historical biological and psychological theories are mostly **speculative**.
- Lifelong and acquired PE share short latency, poor control and negative consequences, but remain **distinct demographic and aetiological populations**.

Lifelong PE	Acquired PE
5-HT_{2C} hyposensitivity and/or 5-HT_{1A} hypersensitivity; neurobiological and genetic variation; maintained and heightened by psychological and environmental factors	Secondary: performance anxiety, psychological or relationship problems, erectile dysfunction (ED) ; occasionally prostatitis, hyperthyroidism, withdrawal from prescribed or recreational drugs
Younger, fewer comorbidities	Older, higher body mass index (BMI) , more hypertension, desire disorder, diabetes mellitus, chronic prostatitis and ED

- PE is more prevalent with **metabolic syndrome**.
- **Circumcision:** two systematic analyses show **no difference** in PE incidence.

Premature Ejaculation and Sexual Performance Anxiety, Psychological or Relationship Problems

- Anxiety is **entrenched in sexual-medicine folklore** as the likeliest cause, with **scant empirical support**.
- Proposed mechanism: anxiety → **sympathetic activation** → earlier emission → lower ejaculatory threshold.
- **Bidirectional with desire:** hypoactive desire may drive PE (unconscious wish to shorten unwanted penetration); chronic frustrating PE may in turn reduce desire.
- **Partner dysfunction** — anorgasmia, hypoactive desire, aversion, arousal disorder, vaginismus — may relate to acquired PE.

Premature Ejaculation and Comorbid Erectile Dysfunction

- As many as **half of men with ED also have PE**.
- **Mechanisms:** need for higher stimulation to achieve an erection → deliberate "**rushing**" to beat **detumescence** of a partial erection → brief latency; compounded by ED-related **performance anxiety**.
- **Caution:** potent men with PE often confuse maintaining an erection **before versus after ejaculation**, answer the **Sexual Health Inventory for Men (SHIM)** inconsistently (especially questions 3 and 4) and receive a **false-positive ED diagnosis**.

Premature Ejaculation and Prostate Disease

- Acute and chronic lower urogenital infection, prostatodynia and chronic prostatitis/chronic pelvic pain syndrome (CP/CPPS) are associated with **ED, PE and painful ejaculation**; PE is the **main sexual symptom in CP/CPPS**.
- Mechanism unknown — inflammation altering sensation and reflex modulation is hypothesised, without convincing evidence.
- **Antibiotic treatment of microbiologically confirmed bacterial prostatitis** in acquired PE → longer IELT and better control.

Premature Ejaculation and Hyperthyroidism

- Most patients with thyroid disorders have sexual dysfunction; PE correlates with **suppressed thyroid-stimulating hormone (TSH)**.
- **Normalising thyroid function markedly reduces PE**.

- Thyroid disease is uncommon in PE clinics → **routine TSH screening is not indicated** unless clinically suggested.

Concept — Lifelong PE is a **primary neurobiological trait**; acquired PE is almost always **secondary to something treatable**, so the first task in acquired PE is to find the cause.

Evaluation of Men Complaining of Premature Ejaculation

Medical History

- **Full medical and sexual history + focused examination + erectile function inventory** + investigations suggested by these.
- **Partner involvement is important but not mandatory**: uninvolved partners may resist change; a cooperative partner builds confidence, skills, self-esteem and control, improving the whole relationship. No controlled PE studies; ED data show interpersonal focus is critical.
- Patients **expect clinicians to ask** but are often embarrassed; asking gives permission and screens associated risks (cardiovascular disease, ED).

TABLE 74.2 RECOMMENDED AND OPTIONAL QUESTIONS TO ESTABLISH THE DIAGNOSIS OF PREMATURE EJACULATION AND DIRECT TREATMENT

Purpose	Questions
Recommended — diagnosis	Time from penetration to ejaculation; can you delay ; are you bothered, annoyed or frustrated
Lifelong vs acquired	When first experienced; since first experience, every/almost every attempt, every partner
Erectile function	Hard enough to penetrate; difficulty maintaining until ejaculation; ever rush intercourse to prevent loss of erection
Relationship impact	Partner upset; partner avoids intercourse; overall relationship affected
Previous treatment	Any prior treatment for PE
Quality of life	Avoid sex from embarrassment; anxious, depressed or embarrassed

From Althof, Abdo et al. (2010)

- The three **recommended questions map exactly onto the three ISSM constructs** (latency, control, consequences); the optional questions serve treatment planning.

Diagnosis of Premature Ejaculation

- **The ISSM unified definition is the basis of diagnosis.**
- Many self-reported sufferers fail ISSM criteria → **variable PE** (occasional, e.g. after long abstinence or with a new partner — a normal variation) or **subjective PE** (normal latency with a false perception of PE, often with reduced control).

Determination of Intravaginal Ejaculatory Latency Time

Patient/partner self-estimate	Stopwatch
Clinical practice ; correlates relatively well with stopwatch; self-report drives treatment-seeking and satisfaction	Trials and observational studies ; objective but intrusive, disrupts pleasure and spontaneity

Patient-Reported Outcome Measures

- **Adjuncts** to history and self-estimated latency; developed mainly as **research tools**; some have good psychometric properties. Five validated questionnaires exist.

COMPARISON — VALIDATED PE QUESTIONNAIRES

Instrument	Validation / database	Clinical use
Premature Ejaculation Profile (PEP)	Extensive; meets most criteria	Yes
Index of Premature Ejaculation (IPE)	Extensive; meets most criteria	Yes
Premature Ejaculation Diagnostic Tool (PEDT)	Brief diagnostic; modest database	Available
Arabic PE Questionnaire; Chinese PE Questionnaire	Minimal validation or trial data	Not recommended

Assessment of Erectile Function

- Assess comorbid ED with a validated instrument — **International Index of Erectile Function (IIEF)** or **IIEF-5 (SHIM)**.
- **Normal erectile function: IIEF erectile function domain ≥ 26 ; IIEF-5 ≥ 22 .**
- ED is **much commoner in men with PE** than without (raised odds across all regions in GSSAB); **more prevalent in acquired than lifelong PE**; PE becomes **more common as ED severity rises**, after age adjustment.

Physical Examination

Lifelong PE	Acquired PE
Diagnosis rests on history — no predictive signs or confirmatory tests; examination highly desirable (lifelong and acquired hard to separate in the young or inexperienced; screens cardiovascular and male-specific disease)	Examination mandatory to identify and relieve the cause

- ED by history or validated instrument; laboratory or imaging studies only as history dictates.
- **Digital rectal examination**, routine in andrology for men **over 40**, helps detect prostatic inflammation or infection.

Concept — PE is a **history diagnosis**: latency, control and bother make it; examination and tests exist to find the cause of acquired PE and the comorbid ED that changes treatment.

Treatment of Premature Ejaculation

- **Subtype-directed**: lifelong → **pharmacotherapy** alone or with graded patient and couple psychosexual therapy; acquired → **aetiology-specific** treatment (psychosexual counselling, ED pharmacotherapy) ± PE pharmacotherapy; variable and subjective → **psychosexual education** and graded psychotherapy.
- **Lifestyle**: PE is associated with ED, metabolic syndrome, sedentary lifestyle, alcohol and BMI → counsel **more exercise and better overall lifestyle**.

Figure 74.3 — Algorithm for the Office Management of PE

Patient/partner history → PE confirmed → secondary to ED or other sexual dysfunction? **Yes** → **manage the primary cause**. No → classify:

- **Variable/subjective** → reassurance, education, behavioural therapy.
- **Acquired** → behavioural therapy, selective serotonin reuptake inhibitor (SSRI) pharmacotherapy or combination → **attempt graduated drug withdrawal after 6–8 weeks**.
- **Lifelong** → SSRI pharmacotherapy, behavioural therapy or combination, chosen by **patient preference**.

Psychosexual Therapy

- **Every man seeking treatment should receive basic psychosexual education or coaching**: prevalence and population latency data to dispel myths; enjoyable activities to broaden the couple's repertoire; strategies for sexual avoidance and reluctance to discuss sex.
- **Aim**: confidence to try medical treatment, less performance anxiety, modified maladaptive sexual scripts.

Pharmacological Treatment

- **Options:** topical local anaesthetics, SSRIs, tramadol, phosphodiesterase type 5 inhibitors (PDE5Is), α -adrenergic blockers.
- **Topical anaesthetics** (lidocaine, prilocaine, benzocaine, alone or combined) to reduce glans sensitivity are the **oldest** pharmacological treatment (1943).
- **SSRIs** (paroxetine, sertraline, fluoxetine, citalopram) and the **tricyclic antidepressant (TCA) clomipramine revolutionised treatment:** block 5-HT transporter reuptake → more synaptic 5-HT → postsynaptic 5-HT receptor stimulation.

Treatment With SSRIs and TCAs

On demand	Daily
Dapoxetine; off-label clomipramine, paroxetine, sertraline, fluoxetine	Off-label paroxetine , clomipramine, sertraline, fluoxetine, citalopram

Dapoxetine

- **Approved in over 50 countries; not approved by the US Food and Drug Administration (FDA).**
- **Rapid-acting SSRI with a short half-life** → suited to on-demand use.
- **No drug–drug interactions reported**, including with PDE5Is.
- **30 or 60 mg, 1–2 h before intercourse** (trial dosing; Table 74.3 gives 1–3 h): better than placebo **from the first dose** — longer IELT, more control, less distress, more satisfaction (randomised controlled trials, RCTs).
- **Equally effective in lifelong and acquired PE;** effective and tolerated in PE with ED on PDE5Is.
- **Adverse effects** uncommon and dose-dependent: nausea, diarrhoea, headache, dizziness; fewer than with daily SSRIs.
- **No increased suicidality;** little withdrawal on abrupt cessation.
- **Real-world discontinuation is very high** within a year — **cost and loss of spontaneity;** satisfaction tracks response.

Off-Label SSRIs and TCAs

- **Daily dosing** of paroxetine, clomipramine, sertraline, fluoxetine or citalopram usually delays ejaculation (doses in Table 74.3).
- **Paroxetine exerts the strongest delay** (meta-analysis).
- **Onset 5–10 days; full effect 2–3 weeks;** sustained long-term.
- **Adverse effects** usually minor, begin in week 1, fade over 2–3 weeks: fatigue, yawning, mild nausea, diarrhoea, sweating. Reduced libido and ED anecdotally less frequent in non-depressed men with PE.
- **Agitation and hypomania** in a few → **avoid SSRIs with a history of bipolar depression.**
- **Platelet serotonin** matters for haemostasis → upper gastrointestinal bleeding risk, especially with **aspirin or non-steroidal anti-inflammatory drugs (NSAIDs).**
- **Priapism** — rare; urgent treatment.
- Long-term: **weight gain** and increased **type 2 diabetes** risk.
- **Semen:** sperm DNA fragmentation and altered semen parameters → possible fertility impact; relevant because most treatment-seekers are in their **mid-30s.**
- **Suicidality:** a small increase in **youth, not adults**, in depression/anxiety trials; **not seen in non-depressed men with PE** → caution in **under-18s** and in **comorbid depression**, especially with suicidal ideation.
- **Avoid abrupt cessation or rapid reduction** of daily SSRIs → SSRI withdrawal syndrome.

- **On-demand off-label SSRIs/clomipramine 3–6 h before:** modestly effective, well tolerated, but **substantially less delay than daily**; can be combined with an initial daily trial or concomitant low-dose daily treatment.
- **Reluctance and dropout** are common — not wanting an antidepressant, effect below expectation, cost.
- **Choosing** is individual (see pair below); where no drug is approved and regulators discourage off-label prescribing, treatment is complicated.

Topical Local Anaesthetics

- **Lidocaine and/or prilocaine** cream, gel or spray — well established, **moderately effective**.
- **Mechanism:** reduced glans sensitivity may inhibit the spinal ejaculatory reflex arc.
- **Drawbacks:** penile hypoaesthesia; **transvaginal absorption** → vaginal numbness and female anorgasmia **unless a condom is used**.
- **Fortacin:** aerosolised **eutectic lidocaine–prilocaine metered spray**; penetrates the **glans mucosa but not keratinised shaft skin**; **European Medicines Agency approval 2014**.
- **Two phase III placebo-controlled trials:** applied **at least 5 min before** intercourse → marked IELT increase with better control, satisfaction and distress scores; **minimal hypoaesthesia or partner transfer**.
- **ISSM: level 1a evidence** for off-label on-demand topical anaesthetics in lifelong PE.
- The **majority stop** effective topical treatment (loss of spontaneity); long-term data scarce.

PDE5 Inhibitors

- **EAU recommends** off-label on-demand or daily PDE5Is for PE **with or without ED**, alone or with PE pharmacotherapy.
- **Rationale speculative:** the NO/cyclic guanosine monophosphate (cGMP) system mediates inhibitory NANC nitrenergic transmission in the urogenital tract.
- Early systematic reviews: **no robust evidence except PE with comorbid ED**.
- **Well-designed RCT of on-demand vardenafil** in potent men with lifelong PE: large rise in geometric mean IELT and better Index of Premature Ejaculation domains (control, confidence, satisfaction, distress).
- **Meta-analyses of RCTs:** PDE5I > placebo; **PDE5I + SSRI > SSRI alone**.
- Current guidelines therefore recommend PDE5Is **with or without ED, alone or with SSRIs**.
- **Online text only:** tramadol, α 1-adrenoceptor antagonists, intracavernosal injection, investigational drugs, acupuncture, surgical neurotomy, cryoablation and dorsal penile nerve neuromodulation.

On-demand dapoxetine	Daily off-label SSRI
Approved (not FDA); short half-life; fewer side effects; little withdrawal; suits infrequent intercourse	Off-label; greater delay (paroxetine strongest); full effect 2–3 weeks; must not stop abruptly; suits established relationships

TABLE 74.3 DRUG THERAPY FOR PREMATURE EJACULATION

Drug	Dose	Dosing	Comment	Evidence
Dapoxetine	30–60 mg	On demand, 1–3 h before	Approved in >50 countries	High
Paroxetine	10–40 mg	Daily	Strongest delay	High
Sertraline	50–200 mg	Daily		High
Fluoxetine	20–40 mg	Daily		High
Citalopram	20–40 mg	Daily		High

Drug	Dose	Dosing	Comment	Evidence
Clomipramine	12.5–50 mg	Daily, or on demand 3–4 h before	TCA	High
Tramadol	25–50 mg	On demand, 3–4 h before	Opiate addiction risk	Low
Topical lidocaine/ prilocaine	Patient-titrated	On demand, 20–30 min before	Condom advised	High
Alprostadil	5–20 µg	Intracavernosal, 5 min before	Priapism, corporal fibrosis	Very low
PDE5Is	Sildenafil 25–100 mg; tadalafil 10–20 mg; vardenafil 10–20 mg	On demand, 30–50 min before	Normal erectile function: low. With ED (possibly better with SSRI): moderate	Low / moderate

All agents listed for lifelong and acquired PE.

- The table ranks options by **evidence**: SSRIs, clomipramine and topical anaesthetics high; PDE5Is depend on **erectile status**; tramadol low; intracavernosal alprostadil very low.
- **Pattern of timing**: dapoxetine 1–3 h; clomipramine and tramadol 3–4 h; PDE5Is 30–50 min; topicals 20–30 min (Fortacin ≥5 min); intracavernosal 5 min.
- **Trap**: only dapoxetine (and in Europe Fortacin) is licensed for PE; everything else is off-label.

KEY POINTS ON PREMATURE EJACULATION

- PE is a common sexual dysfunction.
- PE carries negative psychological consequences — distress, bother, frustration — affecting quality of life (QoL), relationships, self-esteem and self-confidence, and can obstruct single men forming new relationships.
- The evidence-based ISSM definition of lifelong and acquired PE should form the basis of office diagnosis.
- Limited evidence suggests lifelong PE has a genetic basis; acquired PE is most often due to performance anxiety, psychological or relationship problems and/or ED.
- Oral SSRIs, topical anaesthetics and PDE5Is are effective and safe.
- Psychosexual cognitive behavioural therapy (CBT) has a limited first-line role in lifelong PE but an important adjunctive role, especially in acquired PE from performance anxiety.
- Acquired PE secondary to ED, hyperthyroidism, chronic lower urogenital infection, prostatodynia or CPPS needs etiology-specific treatment alone or with an SSRI.

Concept — PE pharmacotherapy is **serotonergic at its core**; everything else is peripheral desensitisation (topicals) or treatment of a co-driver (PDE5Is for ED), and the practical limiter is **adherence, not efficacy**.

DELAYED EJACULATION, ANEJACULATION AND ANORGASMIA

- Any disease, drug or operation that interferes with **central control**, the **sympathetic supply to the vas and bladder neck**, the **somatic efferent supply to the pelvic floor** or the **somatic afferent supply from the penis** can cause DE, anejaculation, RE and/or anorgasmia — the causes are **manifold**.

Definition, Terminology and Characteristics of Men With Delayed Ejaculation

- DE, RE and inhibited ejaculation (IE) are probably the **least common, least studied, least reported and least understood** male sexual dysfunctions — yet they deny fulfilment to both partners, worse when **procreation** is a goal.
- **Range:** varying delay → complete inability; reduced volume, force and sensation may accompany.
- **Extremes:** **anejaculation (time)** and **RE (direction)**; **DE is commoner**.
- **Anorgasmia:** perceived absence of orgasm or sexual anhedonia, **independent of whether the physiological events of ejaculation occur**.

Terminology and Definition

- Synonyms used interchangeably: DE, inadequate ejaculation, IE, idiopathic anejaculation, primary impotentia ejaculationis, psychogenic anejaculation. **IE**, if distinguished, means **complete absence** (no consensus). **DE is the preferred umbrella term** for any delay or absence.

COMPARISON — DEFINITIONS OF DE

	DSM-5 (2013)	Third International Consultation on Sexual Dysfunction	ICD-11 (2018)
Core	Marked delay or absence of ejaculation	Persistent/recurrent difficulty, delay or absence of orgasm after sufficient stimulation	Inability or excessive latency despite adequate stimulation and desire
Frequency	~75–100% of partnered occasions; not desired	Persistent or recurrent	—
Distress	Clinically significant	Personal distress	—
Exclusions / specifiers	Not a non-sexual mental disorder, severe relationship distress, substance or medical cause; lifelong/acquired, generalised/situational, mild/moderate/severe	—	—

ICD-11, *International Classification of Diseases, 11th revision*

- **No operationalised latency criterion** exists. Most functional men ejaculate **4–10 min** after intromission; two standard deviations above the mean ≈ **21–23 min** → a clinician may regard **>25–30 min with distress**, or stopping from exhaustion or irritation, as DE; symptoms plus help-seeking usually suffice.
- Survey data: **latency and control distinguish DE**; bother/distress does not.

The Epidemiology of Delayed Ejaculation

- Prevalence **unclear**: historically low; recent samples higher; adding "lacking the ability to ejaculate sooner" returns it to the historical low level.
- In older men (50–79, 2003 multinational survey) some form of **EjD is as prevalent as ED**.
- **Rises with age** (moderate positive relation; climbs across successive age bands); varies by race; associated with **ED, lower urinary tract symptoms (LUTS), low testosterone, hyperlipidaemia**. Limitation: studies do not subtype EjD or use validated scales.
- **Lifelong versus acquired**: population prevalence ~1% versus 4%; of DE cases ~one-quarter lifelong, three-quarters acquired.
- Global or situational. **Coital anejaculation** often drives presentation — especially very religious men referred for **infertility**; others present distressed by failure with manual, oral or vaginal partner stimulation.

- Most men with acquired DE **can masturbate to orgasm**; some will not or cannot; masturbatory capacity may be lost after emotional or physical trauma.
- Men with DE: high relationship distress, dissatisfaction, performance anxiety and health concerns; **lower coital frequency**.
- **Distinguishing feature: normal erections**, sometimes prolonged, with **low subjective arousal**.

Aetiology of Delayed Ejaculation/Anejaculation

- Lifelong or acquired, global or situational; complex, multiple pathophysiologies.
- **Psychological factors**: performance anxiety, relationship conflict, poor sexual communication, hypoactive desire, fear of fathering, fear of harm to self or partner, religious shame; **idiosyncratic solitary masturbation** that cannot be replicated in coitus conditions DE.
- **Commonest in practice**: psychogenic IE; **age-related degeneration of penile afferents and Pacinian corpuscles**; hypogonadism; **diabetic autonomic neuropathy**; SSRIs and major tranquillisers; radical prostatectomy or major pelvic surgery; radiotherapy.

TABLE 74.4 CAUSES OF RETROGRADE EJACULATION, DELAYED EJACULATION, ANEJACULATION AND ANORGASMIA

Category	Causes
Ageing	Degeneration of penile afferent nerves
Psychogenic	Inhibited ejaculation
Congenital	Müllerian duct cyst; Wolffian duct abnormality; prune belly syndrome
Anatomical	Transurethral resection of the prostate (TURP); bladder neck incision
Neurogenic — disease	Diabetic autonomic neuropathy; multiple sclerosis; spinal cord injury
Neurogenic — surgery	Radical prostatectomy; proctocolectomy; bilateral sympathectomy; abdominal aortic aneurysmectomy; para-aortic lymphadenectomy
Infective	Urethritis; genitourinary tuberculosis; schistosomiasis
Endocrine	Hypogonadism; hypothyroidism
Medication	α -methyl dopa; thiazides; tricyclic and SSRI antidepressants; phenothiazines; alcohol abuse

- The table is a **map of the reflex arc**: afferent (ageing), central (psychogenic, drugs, endocrine), efferent sympathetic (neuropathy, retroperitoneal and pelvic surgery), bladder neck (TURP, incision) and ducts (congenital, infective).
- **Trap**: the anatomical causes produce **RE, not DE** — the table deliberately pools all four disorders.

Psychogenic Delayed Ejaculation

- Usually **performance anxiety** diverting attention from erotic cues. Psychodynamic theories (fear; anxiety from repression, trauma, abuse or prior failure; hostility; religious orthodoxy; relationship difficulty) have **no well-controlled support**.
- **AUA**: refer men with lifelong or acquired DE to a mental health professional with sexual health expertise.
- **Religious orthodoxy**: limited sexual learning or inhibition; minimal or no masturbation; guilt about "spilling seed" → idiosyncratic masturbation; little premarital contact with women.
- **Idiosyncratic, vigorous masturbation** or an **autosexual orientation** (more arousal from masturbation than intercourse) preconditions acquired DE.

- Such men get **erections without subjective arousal**, falsely taken by both partners as readiness for orgasm; **fantasy–reality disparity** further inhibits arousal.
- Advice: **modify positions and practices** to raise arousal.

Congenital Disorders

- **Müllerian duct obstruction** (incomplete absorption of remnants); **Wolffian duct abnormalities** affecting vas, ejaculatory duct and seminal vesicle; **prune belly syndrome**.

Infectious Disorders

- **Sexually transmitted infection** (gonorrhoea, non-specific urethritis) → scarring and obstruction anywhere in the tract, especially if treatment is delayed.
- **Urinary infection with epididymitis** → obstruction, possibly at ejaculatory duct level.
- **Schistosomiasis** (endemic Africa; returning water-sports tourists) → haemospermia; fibrosis and calcification → obstruction.
- **Genitourinary tuberculosis** → extensive damage; healing calcification may make lesions irreparable.
- No definitive evidence of the actual frequency of DE in these infections.

Ejaculatory Duct Obstruction

- **Congenital** (Müllerian duct cysts) or **acquired** (prostatitis, chlamydia and other infection, calculus); causes **1–5% of male infertility**.
- **Bilateral**: very **low-volume, low-pH, low-fructose, low-viscosity azoospermic** semen, or aspermia — the fructose-rich seminal vesicle fluid (**~80% of volume**) is missing.
- **Unilateral or partial**: pelvic pain and oligospermia.
- **Imaging** — transrectal ultrasound (TRUS), magnetic resonance imaging (MRI), seminal vesicle aspiration — has **relatively low sensitivity**.

COMPARISON — LOW OR ABSENT EJACULATE

	Bilateral ejaculatory duct obstruction	Bilateral vasal obstruction	RE	Failure of emission
Volume	Very low	Normal	Low/absent antegrade	Absent
pH, fructose	Low	Normal	—	—
Sperm	Azoo-/oligospermia	Azoospermia	Sperm and fructose in post-orgasm urine	None in urine
Orgasm	Present; pelvic floor contracts; often post-orgasm pain	Present	Usually present	Variable

Endocrinopathy

- **Hypothyroidism** → DE; **hyperthyroidism** → PE.
- **Hypogonadism and low testosterone** are associated with DE or anejaculation; contrary data found **no IELT–testosterone relation** in self-reported DE — the relation is complex and DE multifactorial.
- **AUA**: **normalise testosterone** in DE with concomitant deficiency.
- **Hyperprolactinaemia** inhibits hypothalamic gonadotrophin-releasing hormone (GnRH) → low testosterone, low desire, ED and DE; prolactin may act through serotonergic pathways.

Iatrogenic Causes

- Any prescribed or recreational drug altering **serotonin, dopamine or oxytocin** may change latency; DE is notable in **alcohol dependence**.

- **SSRIs**: sexual dysfunction very common, **EjD the commonest form**; DE risk roughly **sevenfold**.
- **Antipsychotics** → DE (dopamine antagonism and/or raised prolactin) and **RE (α -adrenergic antagonism at the bladder neck)**.
- **Management**: alternative drug, dose adjustment or staged cessation.

Treatment of Male Pelvic Cancers

- Sexual function and QoL are now central outcomes; modern surgery, chemotherapy and radiotherapy spare more function.

Prostate cancer

- Commonest non-skin malignancy in Western men; external beam radiotherapy (EBRT), brachytherapy and nerve-sparing open or robotic radical prostatectomy (RP/RALP) are the common curative treatments.
- **After RP/RALP: no ejaculation, but orgasm preserved** (less or more intense); **arousal incontinence** and **climacturia** (urinary incontinence at orgasm).
- **Climacturia is less frequent after EBRT** than after RP; decreased orgasmic sensation occurs after both.
- After RALP: anorgasmia, reduced orgasm intensity and climacturia all occur; **nerve-sparing reduces anorgasmia; urinary incontinence increases climacturia**.
- **Radiotherapy** (reports since the 1980s): falls in desire and orgasm frequency; **reduced ejaculate volume near-universal after EBRT**; reduced ability to ejaculate; anejaculation, anorgasmia, weaker orgasm and longer time to orgasm; incontinence during sex uncommon.
- **Favourable prognostic factors after radiotherapy**: younger age and higher intercourse frequency.
- **Brachytherapy**: dry ejaculation (early series — all had prior TURP); **discomfort with ejaculation** from prostatic oedema reducing urethral elasticity, sometimes persisting 18–24 months; reduced libido.
- Testosterone is low 3–8 years after EBRT (lower in older men), but the **testicular dose is only a small fraction** of that affecting androgen production; **spermatogenesis is more radiosensitive than androgen production**.
- **Prior TURP disrupts bladder neck closure → RE**, explaining ejaculatory disturbance after radiotherapy in those patients.

Rectal carcinoma

- EjD after treatment is common: **anejaculation, RE and painful ejaculation**.
- **Preoperative radiotherapy** reduces local relapse and may improve survival; with **total mesorectal excision (TME)** it is common in Europe.
- **Sharp mesorectal dissection with visualisation and preservation of pelvic autonomic nerves** → excellent erectile and ejaculatory results.
- Preoperative radiotherapy may impair sexual function (single small study; inconclusive).

Testicular cancer

- Germ cell tumours **~1% of male cancers**; long-term survival in early disease approaches **100%**; young, fertile men → ejaculation matters.
- Retroperitoneal lymph node dissection (RPLND) for post-chemotherapy residual nonseminoma is better documented than abdominal radiotherapy for seminoma.

Non-nerve-sparing RPLND	Nerve-sparing RPLND
Anejaculation in the majority; RPLND an independent predictor of EjD	Antegrade ejaculation preserved in most (open and robot-assisted); libido and orgasm normal

- **Orchidectomy + abdominal radiotherapy**: lower erection, libido and orgasm scores than controls; **ability to ejaculate unchanged but semen volume noticeably reduced**.

- Ejaculatory and orgasmic disturbance **higher after radiotherapy than surveillance** (not significant); declines correlate with age.
- PE reported by up to half — but **the same as before treatment**.
- **Superior hypogastric plexus**: sympathetic, **mediates emission**; fenestrated network **anterior to the lower abdominal aorta**; hypogastric nerves leave its inferior pole bilaterally and connect with S1–S2 roots. Hard to see at RPLND → low volume or dry ejaculation. Lies in radiotherapy fields for rectal and prostate cancer, though dose alone does not explain the dysfunction.
- **Orgasm is more complex than ejaculation** — cortical input.

Neurological Disorders

- **Age-related degeneration of fast-conducting penile afferents and Pacinian corpuscles**, diabetic autonomic neuropathy, multiple sclerosis and spinal cord injury (SCI) are often associated with DE/anejaculation — most frequent in **SCI**, then diabetes, then multiple sclerosis.

Spinal Cord Injury

- Ejaculation is severely impaired; **Bors and Comarr** linked outcome to **level and completeness**.

Erection	Ejaculation
Reflexogenic erection preserved with upper motor neuron (UMN) lesions, lost with complete lower motor neuron (LMN) lesions	Improves with descending lesion level — rare with complete UMN, commoner with LMN

TABLE 74.5 CORRELATION OF ERECTION, EJACULATION AND INTERCOURSE WITH LEVEL AND SEVERITY OF SPINAL CORD INJURY

Cord lesion	Reflexogenic erection	Psychogenic erection	Successful coitus	Ejaculation
UMN complete	92%	9%	66%	1%
UMN incomplete	93%	48%	86%	22%
LMN complete	0%	24%	33%	15%
LMN incomplete	0%	1%	100%	100%

From Comarr (1970) — landmark data

- Text gives complete UMN as **under 5%**; LMN ejaculation depends on an **intact thoracolumbar sympathetic outflow**. When ejaculation occurs, **orgasm may be absent and RE is common**.
- **Trap**: reflexogenic erection needs an intact sacral arc (UMN lesion); ejaculation needs the thoracolumbar and sacral outflow intact (lower, incomplete lesion) — the two move in **opposite directions** with level.
- **Semen retrieval in SCI**:
Penile vibratory stimulation — successful in a large majority (up to ~70%) → **electroejaculation** (electrical stimulation of efferent sympathetic hypogastric plexus fibres) — effective and safe → **prostate massage** (alternative; place in the algorithm unsettled) → **surgical sperm retrieval** — testicular sperm extraction (TESE), testicular sperm aspiration (TESA), microsurgical TESE, microsurgical epididymal sperm aspiration (MESA), percutaneous epididymal sperm aspiration (PESA), vasal aspiration; TESE succeeds in most.
- **Bladder preparation** (empty, instil sperm-wash medium) because of RE risk.
- Vibratory stimulation and electroejaculation both risk **autonomic dysreflexia** → **pretreat with nifedipine**.
- **Semen quality** initially senescent — low count and motility — but **improves with repeated ejaculation**. Causes: chronic urinary infection, dilution with urine, medications, raised scrotal temperature from prolonged sitting, stasis of prostatic fluid.

- **Testicular histology:** hypospermatogenesis, maturation arrest, tubular atrophy, germ cell hypoplasia, interstitial fibrosis, Leydig cell hyperplasia; catheter-related prostatitis, epididymitis and epididymo-orchitis add obstruction and testicular damage.
- **Incomplete lesions** → better sperm density and motility; paternity (natural or assisted reproductive technology, ART) achievable in a substantial proportion.

Evaluation of Men With Delayed Ejaculation

- **Full medical and sexual history, focused examination, serum testosterone**, plus investigations suggested by these.
- **History is key.** First **exclude:** RE, absent puberty, genital tract obstruction, anorgasmia and mimics — ED, subtly low libido, ejaculatory pain, partner dysfunction, orientation conflict, paraphilic interests.
- **Detail:** ejaculatory response, sensation, frequency, techniques; cultural context and history; the whole cycle (desire, arousal, ejaculation, orgasm, refractory period); partner's view and any partner dysfunction; overall relationship satisfaction.
- Then **lifelong or acquired, global or situational** (Table 74.6); if ejaculation fails, **duration of thrusting and reason for stopping** (fatigue, detumescence, futility, partner request).
- **Premonitory sensation** during intercourse or masturbation → arousal close to threshold. Note what improves or worsens performance and his ability to relax, sustain and heighten arousal and concentrate.
- Establish **negative consequences; frequency of intercourse and who initiates** are useful surrogates; explore the non-sexual relationship.
- **Acquired DE:** review illness, surgery, drugs and life events (e.g. fear of hurting a wife after mastectomy → partial arousal); societal and religious attitudes ("spilling of seed as a sin").

TABLE 74.6 RECOMMENDED AND OPTIONAL QUESTIONS TO ESTABLISH THE DIAGNOSIS OF DELAYED EJACULATION AND DIRECT TREATMENT

Purpose	Questions
Recommended — diagnosis	How often can you ejaculate in intercourse; time to ejaculate or stop ; how often close to ejaculation; why you stop ; ever feel you ejaculated but no semen ; bothered/frustrated; how often you ejaculate during masturbation
Lifelong vs acquired	When first experienced; since first experience, every attempt, every partner
Erectile function	Hard enough to penetrate; difficulty maintaining
Relationship impact	Partner upset; either partner avoids intercourse; overall relationship affected
Previous treatment; QoL	Any prior treatment; anxious, depressed or embarrassed

- **Pattern:** DE needs more recommended questions than PE because the **failure mode** (why he stops) and the **masturbation response** separate coital-specific from global DE; "ejaculated but no semen" screens for **RE or obstruction**.
- **Examination:** testes, epididymides, presence of vasa; **screening morning total testosterone**; further hormones or imaging as indicated.
- **Digital rectal examination** (prostate size, anal tone, **bulbocavernosus reflex**) in most men — except young men with situational, clearly psychogenic IE.
- **Neuropathy suspected** → pudendal somatosensory and motor evoked potentials, sacral reflex arc testing, sympathetic skin responses.
- Utility of further tests unclear; **morning testosterone**; electrolytes, lipids, glycated haemoglobin (HbA1c) for neuropathic or vascular predisposition.

- **Orgasm with absent or low-volume antegrade ejaculate → RE or ejaculatory duct obstruction.** Sperm in post-masturbation first-void urine → **RE**; azoo-/oligospermic, low-viscosity, low-fructose, low-pH semen → **obstruction**.
- Aetiology unclear → culture of expressed prostatic secretion and urine, urine cytology, prostate-specific antigen (PSA); TRUS, MRI or scrotal ultrasound for local disease.

Figure 74.4 — Algorithm for the Office Management of DE

Is there orgasm?

- **Never** → with nocturnal or masturbatory emissions → **inhibited male orgasm** → psychosexual therapy.
- **Sometimes** → inhibited male orgasm → psychosexual therapy; or **age-related degeneration** → reassure, alter sexual technique.
- **Always** → **is there ejaculation?** No → **sperm in urine after orgasm?**
 - Yes → **RE** → reassure/educate, pharmacotherapy, surgery.
 - No → **aspermia** → **failure of emission** (neurogenic, metabolic, drug) → disease-specific management; or **ejaculatory duct obstruction**.

Treatment of Men With Delayed Ejaculation/Anejaculation

- **Aetiology-specific**; address **infertility** in men of reproductive age; psychoeducation and/or psychosexual therapy, pharmacotherapy or integrated treatment.
- **Before pelvic surgery**, counsel reproductive-age couples about anejaculatory infertility and the availability of **sperm harvesting and ART**.
- **Behavioural**: low risk — alternative practices, scripts, sexual enhancement devices; at minimum, open discussion of needs.
- **Lifestyle**: more time together for intimacy, minimal alcohol, sex when not tired, maximal penile stimulation, **pelvic floor training**.
- **Drugs** — **limited success**, all off-label: **central dopaminergic, anti-serotonergic or oxytocinergic**, or **peripheral adrenergic**. **None approved**; most have limited efficacy, significant side effects or are experimental. Choice by aetiology, illness features, preference, evidence and physician familiarity.
- **Survey of sexual health providers**: overall treatment success modest (about 40%); **cabergoline, bupropion and oxytocin** most used. Drugs rarely restore normal ejaculation but may give partial antegrade ejaculation or RE; **poor results in psychogenic and neuropathic DE**.
- **Cabergoline** — central **D2 agonist** (licensed for hyperprolactinaemia): shortens latency; also used in anhedonia and anorgasmia; **escalating dose** (Table 74.7). Adverse effects: nausea, dizziness, rarely **cardiac valvular, pericardial, pulmonary and retroperitoneal fibrosis** → caution long term; **echocardiography every 12 months**. Efficacy independent of age and aetiology.
- **Pseudoephedrine** (α 1-agonist) on demand and **reboxetine** (inhibits synaptic noradrenaline reuptake) — **limited efficacy**.
- **Cyproheptadine** (central serotonin antagonist antihistamine) and loratadine: anecdotal reversal of **SSRI-induced anorgasmia**; given chronically or on demand; **sedation** limits it.
- **Anecdotal only**: amantadine, apomorphine, bromocriptine, bupropion, lisdexamfetamine, buspirone.
- **Intranasal oxytocin** during coitus: case reports positive, but **both RCTs showed no superiority to placebo**. Pharmacokinetics: supraphysiological plasma levels (peak ~15 min) but **little enters cerebrospinal fluid** (peak up to 75 min), with no plasma-CSF correlation. Adverse effects: flushing, headache, strange taste. No conclusion without dose-response studies.
- **Invasive options** — pudendal nerve release, intracavernosal injection, platelet-rich plasma, surgery — have **no peer-reviewed data; not recommended outside an ethics-approved trial**.

- **Neuropathic DE is usually irreversible** → counsel alternative routes to mutual satisfaction.
- **Psychotherapy** targets individual and relationship issues; once organic and drug causes are excluded, refer to an **expert psychosexual therapist**; benefit depends on severity and engagement.

TABLE 74.7 DRUG THERAPY FOR DELAYED EJACULATION AND ANEJACULATION

Drug	Mechanism	As needed	Regular
Cabergoline	Central D2 agonist	—	0.5–2.0 mg every 3 days (text: up to 2–3 mg)
Pseudoephedrine	α1-agonist	120 mg 1–2 h before (text)	60–120 mg 6-hourly the day before, then twice on the day of coitus (table)
Amantadine	—	100–400 mg for 2 days before	100–200 mg twice daily
Reboxetine	Noradrenaline reuptake inhibition	—	4–8 mg daily
Oxytocin	Oxytocinergic	24 IU intranasal during coitus	—
Bupropion	—	—	150 mg once or twice daily
Buspirone	—	—	5–15 mg twice daily
Cyproheptadine	Serotonin antagonist	4–12 mg 3–4 h before (text range 2–16 mg)	—
Lisdexamfetamine	—	30–70 mg 3–4 h before	—

- The table's pattern: every agent either **raises dopamine/noradrenaline or blocks serotonin** — the mirror image of the PE drug table.
- **Trap:** none is licensed for DE; cabergoline needs **annual echocardiography**.

KEY POINTS ON DELAYED EJACULATION

- The causes of DE and anejaculation are manifold.
- Failure of ejaculation can be lifelong (25%) or acquired (75%); global, or intermittent or situational.
- Treatment should be aetiology-specific and address infertility in men of reproductive age.
- Drug treatment of DE or anejaculation has met with limited success.

Concept — DE is failure somewhere along a **long reflex arc** — **afferent, central, efferent or ductal** — so there is no single drug for it; treatment is aetiological, and when the cause is neuropathic or psychogenic results are poor.

RETROGRADE EJACULATION

- **Antegrade ejaculation requires a closed bladder neck** (and proximal urethra); surgery compromising closure causes RE.
- **Transurethral incision of the prostate (TUIP):** risk depends on **one versus two incisions** and whether the incision stays at the bladder neck or **extends to the verumontanum** — contraction of urethral smooth muscle at the verumontanum may help prevent RE.
- **TURP carries a higher RE incidence than TUIP.** Some antegrade ejaculation and orgasmic sensation usually persist, though both decline with age.

COMPARISON — EJACULATORY OUTCOME BY BENIGN PROSTATIC HYPERPLASIA PROCEDURE

Procedure	Ejaculatory outcome
TUIP	RE 5–45%
TURP	RE 42–100%
Modified (ejaculation-preserving) TURP	Antegrade 66–91%
Modified photovaporisation	Antegrade 87–96%
Prostatic urethral lift (PUL)	Superior to TURP in RCT (preserved 100% vs 34%)
Aquablation	Superior to TURP in RCT (preserved 90% vs 34%)

- **Drugs** preventing bladder neck shutdown: antihypertensives, thiazides, **α -blockers**, antipsychotics, antidepressants.
- **Neurological**: SCI, multiple sclerosis, myelodysplasia and — commonest — **uncontrolled diabetes with neuropathy**.
- **Diabetes mellitus**: RE commoner than in age-matched controls; prevalence variable across studies; **not related to diabetes duration, BMI, waist, HbA1c or total testosterone**.
- **Diagnosis**: post-masturbation urine for **sperm and fructose** distinguishes RE from failure of emission; **>5–10 sperm per high-power field** confirms RE; with low-volume ejaculate, **more sperm in urine than in the antegrade ejaculate** = significant RE.

Treatment

- **Bladder neck reconstruction** — results **consistently poor**.
- **Drugs are the most promising**: α -adrenergic sympathetic nerves mediate **both bladder neck closure and emission**.
- **Sympathomimetics** — **pseudoephedrine**, ephedrine, **midodrine**, phenylpropanolamine: release noradrenaline from nerve terminals and may directly stimulate α - and β -receptors → internal sphincter closes → antegrade flow.
- **Pseudoephedrine is the most useful**: **120 mg 6-hourly the day before and twice on the day of coitus** → antegrade ejaculation in a majority of predominantly diabetic men.
- Evidence small; a **double-blind randomised study** (after RPLND): **4 days of pre-treatment** most effective; all α -agonists restored antegrade ejaculation. Systematic review: modest overall efficacy.
- **Adverse effects**: dry mucous membranes, **hypertension**.
- **Imipramine** (TCA, blocks noradrenaline reuptake) **25 mg twice daily**; long-term treatment probably more effective.
- **Antimuscarinics** (brompheniramine, imipramine) **combined with sympathomimetics** raise success over sympathomimetics alone.
- **Endourethral collagen injection** — preliminary data only.

KEY POINTS ON RETROGRADE EJACULATION

- TURP and diabetic autonomic neuropathy are the most common causes of RE.
- RE and failure of emission are distinguished by examining post-masturbatory urine for spermatozoa and fructose.
- Pharmacotherapy — pseudoephedrine, midodrine, imipramine — has variable success.

Concept — RE is a **bladder-neck problem**: any operation, drug or neuropathy that stops the neck closing sends semen backwards, and every treatment tries to make it close again, chiefly by **α -agonism**.

SPERM RETRIEVAL IN MEN WITH ANEJACULATION PURSUING FERTILITY

- Medical treatment of RE may give only **partial antegrade ejaculation**.
- If neither surgery nor drugs achieve antegrade ejaculation → **urinary sperm retrieval** with intrauterine insemination (IUI), in vitro fertilisation (IVF) or intracytoplasmic sperm injection (ICSI) — worthwhile per-cycle pregnancy rates.

ORGASM-ASSOCIATED URINARY INCONTINENCE (CLIMACTURIA)

- **Sexual incontinence after RP** is often overlooked, affects many men and causes significant bother.

Climacturia	Arousal incontinence
Urinary leakage at orgasm	Urinary leakage during foreplay

- **Cause not determined.**
- **Treatment:** void before intercourse → condom → **pelvic floor muscle training (PFMT)** → penile constriction ring → **male urethral sling**.

Ejaculatory Anhedonia

- Also sexual anhedonia or **pleasure dissociative orgasm disorder (PDOD)**; men or women; **aware of orgasm and ejaculation but little or no pleasure**. Little literature; poorly understood.
- **Associations:** depression, fatigue, low libido, physical illness, **hypoactive sexual desire disorder (HSDD)**, hypogonadism, hyperprolactinaemia, SCI, multiple sclerosis, during or after SSRIs or antipsychotics.
- When both ejaculation and orgasm are absent, **separate anorgasmia from absent ejaculation (RE, aspermia)**.

Post-Selective Serotonin Reuptake Inhibitor Sexual Dysfunction

Antidepressant withdrawal syndrome	Post-SSRI sexual dysfunction (PSSD)
After abrupt cessation, interruption or dose reduction of SSRIs or serotonin–noradrenaline reuptake inhibitors (SNRIs); mild and transient — flu-like symptoms, disturbed sleep, movement, mood, cognition	Debilitating ; persists months to years , may resolve spontaneously; no accepted definition; may begin after the first dose , within days–weeks, or persist after stopping

- **PSSD symptoms:** genital anaesthesia, HSDD, ED, PE, ejaculatory anhedonia — **absent before SSRI use**; incidence in one study 5–15%.
- Some cases may really be depression-related HSDD, ED and EjD.
- **Aetiology unknown:** dopamine–serotonin interaction, serotonin neurotoxicity, **5-HT_{1A} receptor down-regulation**.

Orgasmic Headache

- **International Headache Society (IHS):** headache precipitated by sexual activity, **dull and bilateral as excitement rises, suddenly intense at orgasm**; abrupt throbbing or constant headache near orgasm in case series.

Painful Ejaculation

- **Odynorgasmia:** mild discomfort to severe pain during or after ejaculation in the **penis, scrotum and perineum**; poorly characterised.
- **Associations:** urethritis, benign prostatic hyperplasia (BPH), acute or chronic prostatitis, CPPS, seminal vesiculitis, seminal vesicle calculi, ejaculatory duct obstruction; **often no cause found**.

- **Prevalence 1–25%** worldwide; **17–23% in LUTS/BPH** — these men have worse LUTS and bother, more ED, lower ejaculate volume.
- **α -blockers may cause it**; lower incidence with uroselective α 1-blocker **alfuzosin**.
- **Treat the underlying cause**. Surgery if medical therapy fails — TURP, transurethral resection of the ejaculatory ducts (TURED), pudendal neurolysis — **no strong evidence; use with caution**.

Postorgasmic Illness Syndrome

- Recently described, poorly characterised "**orphan**" **disease**: severe myalgia, fatigue and a **flu-like state within 30 min of orgasm**, with debilitating physical and psychological effects.

CONCLUSIONS

- Epidemiological and observational research has clarified PE and its **negative psychosocial effects**; the **multivariate, evidence-based ISSM definition** gives a more discriminating tool and should be the basis of office diagnosis of lifelong PE.
- **Aetiology**: limited evidence for a **genetic basis of lifelong PE**; acquired PE is mostly **performance anxiety, psychological or relationship problems and/or ED**, less often CP/CPPS or hyperthyroidism.
- **Treatment**: psychosexual CBT has a **limited role**; oral SSRIs and topical anaesthetics are effective and safe; **dapoxetine**, despite a modest effect, has a place that market forces will settle once regulatory approval is met. **Tramadol, intracavernosal injection and alternative delivery methods cannot be recommended** pending large well-designed RCTs.
- **DE and anejaculation** become commoner with age, have manifold organic and psychogenic causes, affect both partners and may cause infertility; their treatment is among the **most significant challenges in sexual medicine**, with often disappointing outcomes.